SmartWealth: Goal-Based Investment Strategy for a Digital Wealth Platform

Inter-IIT Tech Meet 15.0 – Practice Problem Statement

1 Business Context

A fictional Indian digital wealth platform, **NiveshNow**, is seeing new retail users invest without clearly connecting investments to goals. Some users choose high-volatility products for near-term expenses; others hold too much idle cash for long-term goals. The platform wants to improve investor outcomes without turning the interface into a trading terminal or a black-box "buy" signal engine.

Your team is the product-and-engineering squad asked to build **SmartWealth**, a prototype that transforms a short investor profile and a small product catalogue into a suitable, cost-aware, and explainable investment plan. The goal is not to predict tomorrow's prices. It is to turn user needs and constraints into defensible investment decisions.

2 Challenge Objectives

The prototype must solve six connected tasks:

1. **Profile and suitability.** Parse a profile (goal, time horizon, risk tolerance, available capital, recurring contribution, emergency-fund position, and liquidity needs). Identify missing or contradictory inputs and request clarification or show a safe warning.
2. **Construct an allocation.** Recommend a portfolio allocation across the supplied fictional product universe. A transparent rules engine is acceptable; constrained optimisation is optional. Avoid selecting products only because they have the highest assumed return.
3. **Check goal feasibility.** Estimate a conservative, base, and optimistic future-value range. Show the goal gap in rupees and one or two possible adjustments (for example, contribution amount or target date). Never present projections as guaranteed returns.
4. **Show cost and liquidity trade-offs.** Calculate the weighted expense ratio, estimated annual cost, and whether the portfolio can meet the profile's liquidity requirement. Flag lock-in or settlement mismatches.
5. **Explain recommendations.** For each allocation, tell the user why it is included, which profile constraint it serves, the key downside, and why at least one tempting alternative was limited or excluded.
6. **Support a what-if decision.** Let a reviewer simulate at least two changes, such as a 15% market shock, a 20% SIP reduction, a shorter goal horizon, or a change in risk tolerance. Explain what changed and why.

3 Input Files Supplied

The three CSV files shipped with this brief are part of the challenge. No external API, paid dataset, or live brokerage account is required.

- `investor_profiles.csv`: 12 synthetic investor personas. Includes goal amount, horizon, existing investment, monthly SIP, risk-tolerance score (1–5), emergency-fund months, liquidity need, experience, and an optional preference.
- `investment_products.csv`: 8 fictional product categories, including risk score, assumed annual return, volatility proxy, annual expense ratio, liquidity days, minimum investment, and lock-in. Values are deliberately simplified.
- `synthetic_monthly_returns.csv`: 36 months of synthetic returns per product (long format). Use it for a stress/consistency check if useful; it is not market history and should not be used to claim real-world performance.

3.1 Minimum Data Validation

Validate that allocations sum to 100% (within 0.1 percentage point), weights are non-negative, required fields are present, all referenced product IDs exist, and projected values are finite. Handle empty/malformed fields explicitly rather than silently filling them with arbitrary values. The data are intentionally small so you can complete the core work in 48 hours.

4 Guardrails and Business Rules

Your method may differ, but the final recommendation must respect these baseline constraints. Explain any additional rules you introduce.

- **Diversification:** No single product may receive more than 40% of a portfolio. A portfolio must include at least two product categories unless the team clearly explains why the provided facts make that unsuitable.
- **Short horizons:** For goals up to 36 months, keep products with risk score 4–5 at or below 10% combined and allocate at least 70% to products with risk score 1–2. For horizons of 37–84 months, use a 30% cap on risk score 4–5 and a 45% minimum in risk score 1–2 products. For horizons over 84 months, use a 55% cap and a 25% minimum, respectively. If the investor's risk tolerance implies a stricter limit, the stricter limit wins.
- **Risk fit:** The weighted portfolio risk score must not exceed the investor's risk-tolerance score by more than 0.25. If a profile cannot be satisfied alongside the other constraints, return a clear exception and a safer fallback instead of hiding the conflict.
- **Liquidity:** Show whether the recommended liquid/short-settlement allocation covers the stated near-term liquidity need. A product with a lock-in that extends beyond the goal date must not be allocated to that goal.
- **Emergency buffer:** If the user reports fewer than 3 months of emergency-fund coverage, flag it prominently and demonstrate a conservative liquid-buffer option before allocating all available capital to a long-horizon goal.
- **Cost transparency:** Report weighted expense ratio and estimated annual product cost using the allocated balance. Do not optimise costs by violating suitability or liquidity constraints.
- **No false certainty:** Use only the provided synthetic assumptions for projection. State that market returns are uncertain and all outputs are educational simulations, not personal financial advice or a guaranteed outcome.

5 Projection Method

For a simple implementation, let P be the initial investment, S the monthly SIP, r the assumed annual portfolio return as a decimal, and n the number of months to the goal. One possible future-value estimate is

$$FV(P, S, r, n) = P(1 + r/12)^n + S\frac{(1 + r/12)^n - 1}{r/12}. \tag{1}$$

Use the limiting form $FV = P + Sn$ when $r = 0$. A weighted return assumption $r_p = \sum_i w_i r_i$ is sufficient for the base estimate, where w_i is a product weight and r_i its supplied annual-return assumption. For a simple scenario range, test $r_p - 3$ percentage points, r_p, and $r_p + 3$ percentage points (with a sensible floor/cap you explain). This is a planning approximation, not a forecast. Teams may use a more defensible method if they explain it.

6 Required User Experience

The reviewer should be able to select a persona and quickly see:

- the proposed allocation (percent and rupee amount), asset-class breakdown, weighted risk score, and blended annual expense ratio;
- the goal projection range and the likely shortfall/surplus against the target;
- two or more plain-language reasons tied directly to profile attributes;
- key warnings: inadequate emergency buffer, liquidity mismatch, short horizon, cost, or goal infeasibility;
- what changes after the reviewer edits a contribution, horizon, or risk tolerance.

A lightweight Streamlit app, React web interface, or well-structured CLI plus a clear demo is acceptable. UI polish helps, but it cannot substitute for correct constraints and calculations. No live prices or real user data should be used.

7 Submission Package

Submit a single repository or ZIP with:

1. **Runnable prototype:** setup instructions, dependency file, and one-command launch instructions.
2. **Recommendation outputs:** a CSV with one row per client/product allocation and a summary CSV with goal projections, risk fit, costs, liquidity status, and warnings.
3. **Short report (maximum 5 pages):** system architecture, allocation logic, assumptions, trade-offs, failure cases, and what you would improve with more time.
4. **Demo (maximum 5 minutes):** show one conservative short-horizon investor, one long-horizon investor, and one edge case or infeasible goal.
5. **Tests:** at least five automated checks, including allocation sums, non-negative weights, lock-in/horizon rules, risk cap, and a malformed-profile case.

8 Evaluation Rubric (100 Points)

- **25 points – Suitability and guardrails:** Recommendations respect horizon, risk, liquidity, emergency buffer, and product constraints; infeasible cases are handled honestly.
- **20 points – Goal planning:** Correct future-value arithmetic, meaningful scenario range, clear gap, and actionable but non-guaranteed adjustment options.
- **15 points – Product/workflow quality:** Reviewer can move from profile to recommendation and what-if in a few clear steps.
- **15 points – Explainability and trust:** User-facing reasons, key downside, cost disclosure, and no unsupported claim of certainty.
- **10 points – Engineering correctness:** Reproducible setup, modular code, validation, readable outputs, and sensible error handling.
- **10 points – Cost and liquidity reasoning:** Blended expenses and liquidity readiness are calculated correctly and affect decisions.
- **5 points – Robustness and communication:** Useful tests, edge-case discussion, concise report and convincing demo.